Problem A. Signed Difficulty

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

You are given a real number $X.Y$, where Y is a single digit.

Print the following: (quotes for clarity)

- X-, if $0 \leq Y \leq 2$;
- X, if $3 \leq Y \leq 6$;
- X+, if $7 \leq Y \leq 9$.

Constraints

- $1 \leq X \leq 15$
- $0 \leq Y \leq 9$
- X and Y are integers.

Input

Input is given from Standard Input in the following format:

$X.Y$

Output

Print the answer.

Sample 1

Input	Output
15.8	15+

Here, printing `15 +` will not be accepted: do not print a space between X and `+`, or between X and `-`.

Sample 2

Input	Output
`1.0`	`1-`

You will not get inputs such as `1.00` and `1`.

Sample 3

Input	Output
`12.5`	`12`

Problem B. Cycle Hit

Time Limit 2000 ms

Problem Statement

You are given four strings S_1, S_2, S_3, and S_4.
Determine whether this sequence of strings has one of each of the following: `H`, `2B`, `3B`, and `HR`.
Here, it is guaranteed that every S_i is `H`, `2B`, `3B`, or `HR`.

Constraints

- Each S_i is `H`, `2B`, `3B`, or `HR`.

Input

Input is given from Standard Input in the following format:

S_1
S_2
S_3
S_4

Output

If the given sequence of strings has one of each of `H`, `2B`, `3B`, and `HR`, print `Yes`.
Otherwise, print `No`.

Sample 1

Input	Output
3B HR 2B H	Yes

We have one of each of `H`, `2B`, `3B`, and `HR`.

Sample 2

Input	Output
2B 3B HR 3B	No

We have no `H`.

Problem C. Power Socket

Time Limit 2000 ms

Problem Statement

Takahashi's house has only one socket.

Takahashi wants to extend it with some number of power strips, each with A sockets, into B or more empty sockets.

One power strip with A sockets can extend one empty socket into A empty sockets.

Find the minimum number of power strips required.

Constraints

- All values in input are integers.
- $2 \leq A \leq 20$
- $1 \leq B \leq 20$

Input

Input is given from Standard Input in the following format:

A B

Output

Print the minimum number of power strips required.

Sample 1

Input	Output
4 10	3

3 power strips, each with 4 sockets, extend the socket into 10 empty sockets.

Sample 2

Input	Output
8 9	2

2 power strips, each with 8 sockets, extend the socket into 15 empty sockets.

Sample 3

Input	Output
8 8	1

Problem D. Rally

Time Limit 2000 ms

Problem Statement

There are N people living on a number line.

The i-th person lives at coordinate X_i.

You are going to hold a meeting that all N people have to attend.

The meeting can be held at any **integer coordinate**. If you choose to hold the meeting at coordinate P, the i-th person will spend $(X_i - P)^2$ points of stamina to attend the meeting.

Find the minimum total points of stamina the N people have to spend.

Constraints

- All values in input are integers.
- $1 \leq N \leq 100$
- $1 \leq X_i \leq 100$

Input

Input is given from Standard Input in the following format:

$$N$$
$$X_1 \ X_2 \ \ldots \ X_N$$

Output

Print the minimum total stamina the N people have to spend.

Sample 1

Input	Output
2 1 4	5

Assume the meeting is held at coordinate 2. In this case, the first person will spend $(1-2)^2$ points of stamina, and the second person will spend $(4-2)^2 = 4$ points of stamina, for a total of 5 points of stamina. This is the minimum total stamina that the 2 people have to spend.

Note that you can hold the meeting only at an integer coordinate.

Sample 2

Input	Output
7 14 14 2 13 56 2 37	2354

Problem E. Can you solve this?

Time Limit 2000 ms

Problem Statement

There are N pieces of source code. The characteristics of the i-th code is represented by M integers $A_{i1}, A_{i2}, ..., A_{iM}$.

Additionally, you are given integers $B_1, B_2, ..., B_M$ and C.

The i-th code correctly solves this problem if and only if $A_{i1}B_1 + A_{i2}B_2 + ... + A_{iM}B_M + C > 0$.

Among the N codes, find the number of codes that correctly solve this problem.

Constraints

- All values in input are integers.
- $1 \leq N, M \leq 20$
- $-100 \leq A_{ij} \leq 100$
- $-100 \leq B_i \leq 100$
- $-100 \leq C \leq 100$

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \ \ M \ \ C \\
B_1 \ \ B_2 \ \ ... \ \ B_M \\
A_{11} \ \ A_{12} \ \ ... \ \ A_{1M} \\
A_{21} \ \ A_{22} \ \ ... \ \ A_{2M} \\
\vdots \\
A_{N1} \ \ A_{N2} \ \ ... \ \ A_{NM}
\end{array}
$$

Output

Print the number of codes among the given N codes that correctly solve this problem.

Sample 1

Input	Output
2 3 -10 1 2 3 3 2 1 1 2 2	1

Only the second code correctly solves this problem, as follows:

- Since $3 \times 1 + 2 \times 2 + 1 \times 3 + (-10) = 0 \leq 0$, the first code does not solve this problem.
- $1 \times 1 + 2 \times 2 + 2 \times 3 + (-10) = 1 > 0$, the second code solves this problem.

Sample 2

Input	Output
5 2 -4 -2 5 100 41 100 40 -3 0 -6 -2 18 -13	2

Sample 3

Input	Output
3 3 0 100 -100 0 0 100 100 100 100 100 -100 100 100	0

All of them are *Wrong Answer*. Except yours.

Problem F. Card Game for Two

Time Limit 2000 ms

Problem Statement

We have N cards. A number a_i is written on the i-th card.
Alice and Bob will play a game using these cards. In this game, Alice and Bob alternately take one card. Alice goes first.
The game ends when all the cards are taken by the two players, and the score of each player is the sum of the numbers written on the cards he/she has taken. When both players take the optimal strategy to maximize their scores, find Alice's score minus Bob's score.

Constraints

- N is an integer between 1 and 100 (inclusive).
- a_i $(1 \leq i \leq N)$ is an integer between 1 and 100 (inclusive).

Input

Input is given from Standard Input in the following format:

N
a_1 a_2 a_3 $\ldots$ a_N

Output

Print Alice's score minus Bob's score when both players take the optimal strategy to maximize their scores.

Sample 1

Input	Output
2 3 1	2

First, Alice will take the card with 3. Then, Bob will take the card with 1. The difference of their scores will be 3 - 1 = 2.

Sample 2

Input	Output
3 2 7 4	5

First, Alice will take the card with 7. Then, Bob will take the card with 4. Lastly, Alice will take the card with 2. The difference of their scores will be 7 - 4 + 2 = 5. The difference of their scores will be 3 - 1 = 2.

Sample 3

Input	Output
4 20 18 2 18	18

Problem G. Break Number

Time Limit 2000 ms

Problem Statement

Takahashi loves numbers divisible by 2.

You are given a positive integer N. Among the integers between 1 and N (inclusive), find the one that can be divisible by 2 for the most number of times. The solution is always unique.

Here, the number of times an integer can be divisible by 2, is how many times the integer can be divided by 2 without remainder.

For example,

- 6 can be divided by 2 once: 6 -> 3.
- 8 can be divided by 2 three times: 8 -> 4 -> 2 -> 1.
- 3 can be divided by 2 zero times.

Constraints

- $1 \leq N \leq 100$

Input

Input is given from Standard Input in the following format:

N

Output

Print the answer.

Sample 1

Input	Output
7	4

4 can be divided by 2 twice, which is the most number of times among 1, 2, ..., 7.

Sample 2

Input	Output
32	32

Sample 3

Input	Output
1	1

Sample 4

Input	Output
100	64

Problem H. Divide the Problems

Time Limit 2000 ms

Problem Statement

Takahashi made N problems for competitive programming. The problems are numbered 1 to N, and the difficulty of Problem i is represented as an integer d_i (the higher, the harder).

He is dividing the problems into two categories by choosing an integer K, as follows:

- A problem with difficulty K or higher will be *for ARCs*.
- A problem with difficulty lower than K will be *for ABCs*.

How many choices of the integer K make the number of problems for ARCs and the number of problems for ABCs the same?

Problem Statement

- $2 \leq N \leq 10^5$
- N is an even number.
- $1 \leq d_i \leq 10^5$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

N
$d_1 \ d_2 \ \ldots \ d_N$

Output

Print the number of choices of the integer K that make the number of problems for ARCs and the number of problems for ABCs the same.

Sample 1

Input	Output
6 9 1 4 4 6 7	2

If we choose $K = 5$ or 6, Problem 1, 5, and 6 will be for ARCs, Problem 2, 3, and 4 will be for ABCs, and the objective is achieved. Thus, the answer is 2.

Sample 2

Input	Output
8 9 1 14 5 5 4 4 14	0

There may be no choice of the integer K that make the number of problems for ARCs and the number of problems for ABCs the same.

Sample 3

Input	Output
14 99592 10342 29105 78532 83018 11639 92015 77204 30914 21912 34519 80835 100000 1	42685

Problem I. Go to School

Time Limit 2000 ms

Problem Statement

Takahashi is a teacher responsible for a class of N students.

The students are given distinct student numbers from 1 to N.

Today, all the students entered the classroom at different times.

According to Takahashi's record, there were A_i students in the classroom when student number i entered the classroom (including student number i).

From these records, reconstruct the order in which the students entered the classroom.

Constraints

- $1 \leq N \leq 10^5$
- $1 \leq A_i \leq N$
- $A_i \neq A_j \ (i \neq j)$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
A_1 \ A_2 \ \ldots \ A_N
\end{array}
$$

Output

Print the student numbers of the students in the order the students entered the classroom.

Sample 1

Input	Output
3 2 3 1	3 1 2

First, student number 3 entered the classroom.

Then, student number 1 entered the classroom.

Finally, student number 2 entered the classroom.

Sample 2

Input	Output
5 1 2 3 4 5	1 2 3 4 5

Sample 3

Input	Output
8 8 2 7 3 4 5 6 1	8 2 4 5 6 7 3 1

Problem J. Alchemist

Time Limit 2000 ms

Problem Statement

You have a pot and N ingredients. Each ingredient has a real number parameter called *value*, and the value of the i-th ingredient $(1 \leq i \leq N)$ is v_i.

When you put two ingredients in the pot, they will vanish and result in the formation of a new ingredient. The value of the new ingredient will be $(x + y)/2$ where x and y are the values of the ingredients consumed, and you can put this ingredient again in the pot.

After you compose ingredients in this way $N - 1$ times, you will end up with one ingredient. Find the maximum possible value of this ingredient.

Constraints

- $2 \leq N \leq 50$
- $1 \leq v_i \leq 1000$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

```
N
v_1 v_2 ... v_N
```

Output

Print a decimal number (or an integer) representing the maximum possible value of the last ingredient remaining.

Your output will be judged correct when its absolute or relative error from the judge's output is at most 10^{-5}.

Sample 1

Input	Output
2 3 4	3.5

If you start with two ingredients, the only choice is to put both of them in the pot. The value of the ingredient resulting from the ingredients of values 3 and 4 is $(3+4)/2 = 3.5$.

Printing `3.50001`, `3.49999`, and so on will also be accepted.

Sample 2

Input	Output
3 500 300 200	375

You start with three ingredients this time, and you can choose what to use in the first composition. There are three possible choices:

- Use the ingredients of values 500 and 300 to produce an ingredient of value $(500+300)/2 = 400$. The next composition will use this ingredient and the ingredient of value 200, resulting in an ingredient of value $(400+200)/2 = 300$.
- Use the ingredients of values 500 and 200 to produce an ingredient of value $(500+200)/2 = 350$. The next composition will use this ingredient and the ingredient of value 300, resulting in an ingredient of value $(350+300)/2 = 325$.
- Use the ingredients of values 300 and 200 to produce an ingredient of value $(300+200)/2 = 250$. The next composition will use this ingredient and the ingredient of value 500, resulting in an ingredient of value $(250+500)/2 = 375$.

Thus, the maximum possible value of the last ingredient remaining is 375.

Printing `375.0` and so on will also be accepted.

Sample 3

Input	Output
5 138 138 138 138 138	138